

Lab02 ME315A

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Questão 1

Número total de voos e número de voos com mais de 10 minutos de atraso na chegada são suficientes para calcular o percentual de voos com mais de 10 minutos de atraso na chegada

#Questão 2

```
#e
getStats <- function(input, pos) {

  #a
  input %>%
    filter(AIRLINE %in% c("AA", "DL", "UA", "US")) %>%

  #b
  filter(!is.na(DAY),
         !is.na(MONTH),
         !is.na(AIRLINE),
         !is.na(ARRIVAL_DELAY)) %>%

  #c
  group_by(MONTH, DAY, AIRLINE) %>%

  #d
  summarise(
    total = n(),
    atrasados = sum(ARRIVAL_DELAY > 10),
    .groups = "drop"
  )
}
```

#Questão 3

```
install.packages("readr")
```

Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.6'
(as 'lib' is unspecified)

```
install.packages("dplyr")
```

Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.6'
(as 'lib' is unspecified)

```
library(readr)  
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
resultado <- read_csv_chunked(  
  "flights.csv",  
  
  #b  
  callback = DataFrameCallback$new(getStats),  
  
  #a  
  chunk_size = 100000,  
  
  #c  
  col_types = cols_only(  
    AIRLINE = col_character(),  
    DAY = col_integer(),  
    MONTH = col_integer(),  
    ARRIVAL_DELAY = col_double()  
  )  
)
```

```
#Questão 4
computeStats <- function(input, pos) {

  input %>%

  #a
  group_by(DAY, MONTH, AIRLINE) %>%
  summarise(
    total = sum(total),
    atrasados = sum(atrasados),
    .groups = "drop"
  ) %>%
  mutate(

    #b
    Cia = AIRLINE,
    Data = as.Date(
      paste(2015, MONTH, DAY, sep = "-")
    ),
    Perc = atrasados / total
  ) %>%
  select(Cia, Data, Perc)
}
```

#Questão 5

```
#a
install.packages("remotes")
```

Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.6'
(as 'lib' is unspecified)

```
remotes::install_github("jayjacobs/ggcal")
```

Skipping install of 'ggcal' from a github remote, the SHA1 (ab1a85ad) has not changed since 1
Use `force = TRUE` to force installation

```
library(ggcal)
library(ggplot2)
```

```

#b
pal <- scale_fill_gradient(
  low = "#4575b4",
  high = "#d73027"
)

#c
baseCalendario <- function(stats, cia) {

  dados <- stats %>%
    filter(Cia == cia)

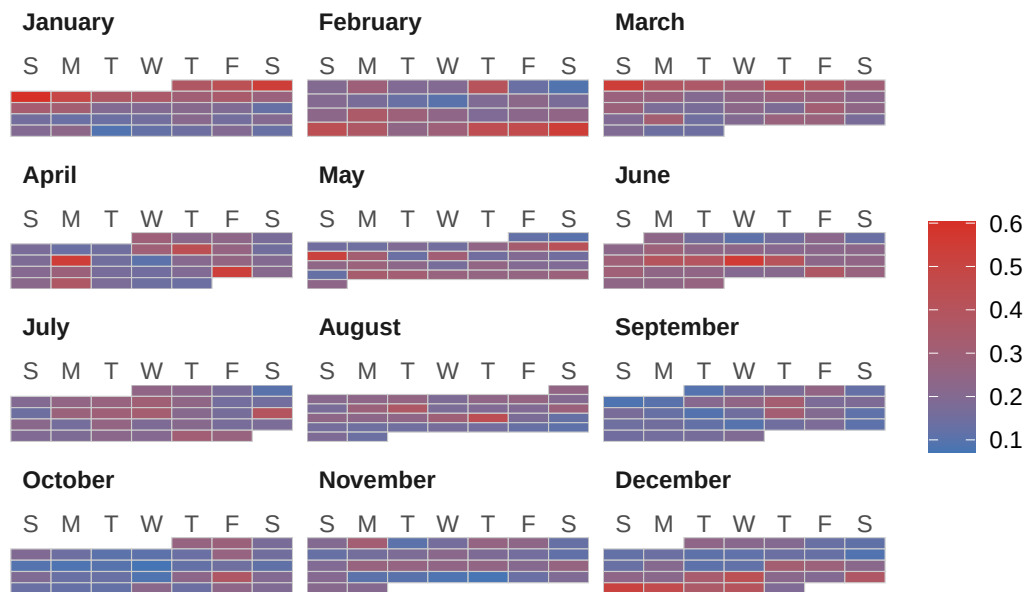
  ggcal(dados$Data, dados$Perc)
}

#d
stats <- computeStats(resultado, 1)
cAA <- baseCalendario(stats, "AA")
cDL <- baseCalendario(stats, "DL")
cUA <- baseCalendario(stats, "UA")
cUS <- baseCalendario(stats, "US")

#e
cAA + pal + ggtitle("Percentual de atrasos - AA")

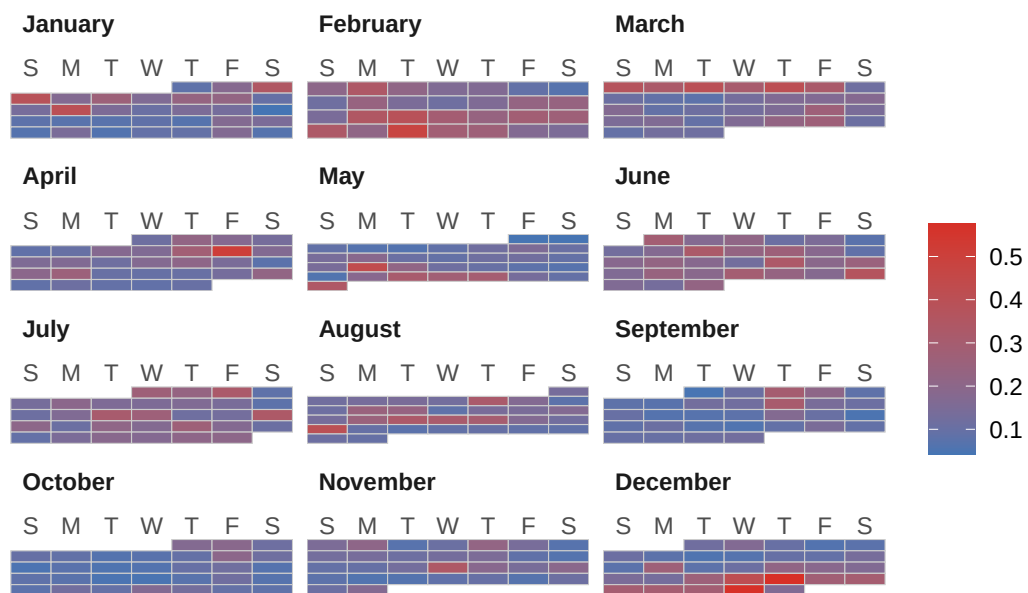
```

Percentual de atrasos – AA



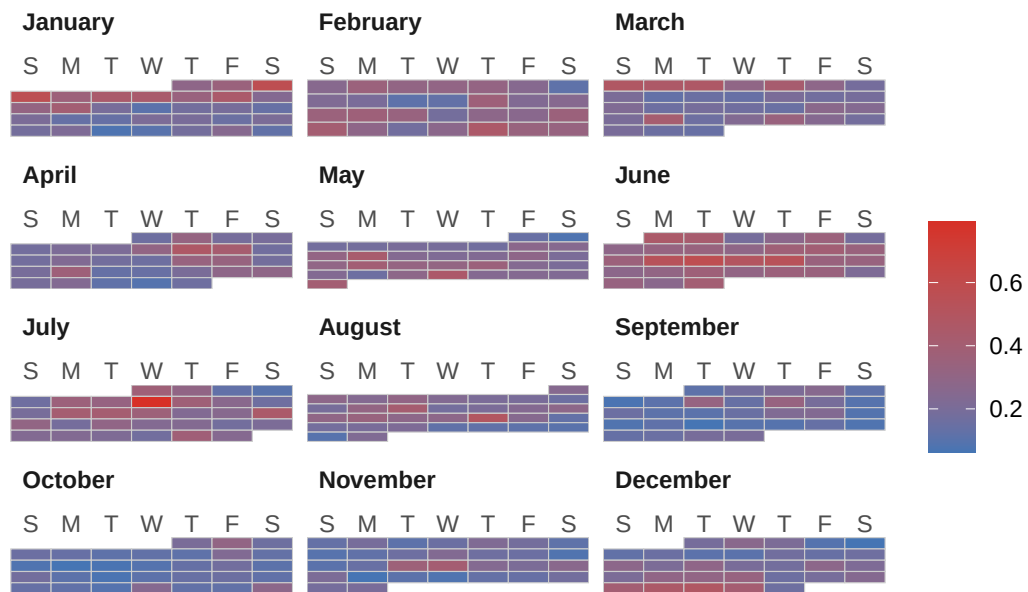
```
cDL + pal + ggtitle("Percentual de atrasos - DL")
```

Percentual de atrasos – DL



```
cUA + pal + ggtitle("Percentual de atrasos - UA")
```

Percentual de atrasos – UA



```
cUS + pal + ggtitle("Percentual de atrasos - US")
```

Percentual de atrasos – US

